

MATHEMATICS

TARGET : JEE- Advanced

CAPS-7**DETERMINANTS & SYSTEM OF
EQUATIONS****SCQ (Single Correct Type) :**

1. The number of positive integral solutions (x, y, z) of the equation

$$\begin{vmatrix} x^3 + 1 & x^2y & x^2z \\ xy^2 & y^3 + 1 & y^2z \\ xz^2 & yz^2 & z^3 + 1 \end{vmatrix} = 11 \text{ is } \underline{\hspace{1cm}}$$

- (A) 0 (B) 3 (C) 6 (D) 12

2. If $x = a, y = b, z = c$ is a solution of the system of linear equations $x + 8y + 7z = 0, 9x + 2y + 3z = 0, x + y + z = 0$ such that the point (a, b, c) lies on the plane $x + 2y + z = 6$, then $2a + b + c$ equals :

- (A) -1 (B) 0 (C) 1 (D) 2

3. Let $A = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ such that $|A| = 0$. If a, b, c are distinct, then the sum of the coordinates of

the fixed point through which the line $ax + by + c = 0$ passes is

- (A) 2 (B) 3 (C) 4 (D) 5

4. If a system of the equation $(\alpha + 1)^3 x + (\alpha + 2)^3 y - (\alpha + 3)^3 = 0$ and $(\alpha + 1)x + (\alpha + 2)y - (\alpha + 3) = 0, x + y - 1 = 0$ is consistent, then the value(s) of α is/are

- (A) 1 (B) 0 (C) -3 (D) -2

5. If a, b, c are complex numbers and $z = \begin{vmatrix} 0 & -b & -c \\ \bar{b} & 0 & -a \\ \bar{c} & \bar{a} & 0 \end{vmatrix}$ is

- (A) purely real (B) purely imaginary (C) 0 (D) none of these

6. If $f(x) = \log_{10}x$ and $g(x) = e^{i\pi x}$ and $h(x) = \begin{vmatrix} f(x)g(x) & [f(x)]^{g(x)} & 1 \\ f(x^2)g(x^2) & [f(x^2)]^{g(x^2)} & 0 \\ f(x^3)g(x^3) & [f(x^3)]^{g(x^3)} & 1 \end{vmatrix}$, then the value of

$h(10)$ is

- (A) 0 (B) 2 (C) 1 (D) 4

7. Let S be the set of all real values of 'a' for which the following system of linear equations

$$ax + 2y + 5z = 1$$

$$2x + y + 3z = 1$$

$$3y + 7z = 1$$



is consistent. Then the set S is :

- (A) equal to R (B) equal to $R - \{1\}$ (C) equal to $\{1\}$ (D) an empty set

8. Let α, β, γ are the real roots of the equation $x^3 + ax^2 + bx + c = 0$ ($a, b, c \in R$ and $a \neq 0$).

If the system of equations (in u, v and w) given by

$$\alpha u + \beta v + \gamma w = 0$$

$$\beta u + \gamma v + \alpha w = 0$$

$$\gamma u + \alpha v + \beta w = 0$$

has non-trivial solutions, then a^2 equals

- (A) b (B) $2b$ (C) $3b$ (D) $4b$

9. For a unique value of p and q , the system of equations given by

$$x + y + z = 6$$

$$x + 2y + 3z = 14$$

$$2x + 5y + pz = q$$

has infinitely many solutions, then the value of $(p+q)$ is equal to

- (A) 14 (B) 24 (C) 34 (D) 44

MCQ (One or more than one correct) :

10. Consider the system of equations

$$ax_1 + x_2 + x_3 = 1$$

$$x_1 + ax_2 + x_3 = 1$$

$$x_1 + x_2 + ax_3 = 1$$

then :

- (A) if $a = 2$, then the system has unique solution.
 (B) if $a = 1$, then the system has infinite solution.
 (C) if $a = -2$, then the system has no solution.
 (D) if $a = 2$, then the system has infinite solution.

Comprehension Type Question:

Comprehension # 1

For $\alpha, \beta, \gamma, \theta \in R$. Let

$$A_{\theta}(\alpha, \beta, \gamma) = \begin{vmatrix} \cos(\alpha + \theta) & \sin(\alpha + \theta) & 1 \\ \cos(\beta + \theta) & \sin(\beta + \theta) & 1 \\ \cos(\gamma + \theta) & \sin(\gamma + \theta) & 1 \end{vmatrix}$$

11. If $a = A_{\pi/2}(\alpha, \beta, \gamma)$, $b = A_{\pi/3}(\alpha, \beta, \gamma)$. Which of the following is true

- (A) $a = b$ (B) $a < b$ (C) $a > b$ (D) $2a = b$

12. $A_{\theta}^2 + A_{\phi}^2 - 2(A_{\theta+\phi})^2$ equals

- (A) $-2A_{\theta}A_{\phi}$ (B) $A_{\theta} + A_{\phi}$ (C) $A_{\theta} - A_{\phi}$ (D) None of these

13. If α, β, γ are fixed, then $y = A_x(\alpha, \beta, \gamma)$ represents

- (A) a straight line parallel to x-axis
- (B) a straight line through the origin
- (C) a parabola with vertex at origin
- (D) None of these

Comprehension # 2

Consider the system of equations :

$$ax + 4y + z = 0$$

$$2y + 3z - 1 = 0$$

$$3x - bz + 2 = 0$$

Then

14. The given system of equations will have a unique solution if

- (A) $ab = 15$ (B) $ab \neq 15$ (C) $ab = 5$ (D) $a \neq 5$

15. The system of equations will have infinite solutions if

- (A) $a = 3, b = 2$ (B) $a = 3, b = 4$ (C) $a = 5, b = 3$ (D) $a = 3, b = 5$

16. The given system of the equations will have no solution if

- (A) $ab = 15, a \neq 3$ (B) $ab \neq 15, a \neq 3$ (C) $ab \neq 15, a = 3$ (D) None of these

Numerical based Questions :

17. If $\begin{vmatrix} 1 & x & x^2 \\ x & x^2 & 1 \\ x^2 & 1 & x \end{vmatrix} = 3$ then the value of $\begin{vmatrix} x^3 - 1 & 0 & x - x^4 \\ 0 & x - x^4 & x^3 - 1 \\ x - x^4 & x^3 - 1 & 0 \end{vmatrix}$ is ____.

18. If $D = \begin{vmatrix} 10! & 11! & 12! \\ 11! & 12! & 13! \\ 12! & 13! & 14! \end{vmatrix}$ then find the units digit of $\frac{D}{(10!)^3}$

19. Total number of 2×2 determinants whose entries are from the set $\{-1, 0, 1\}$ has value equal to -1 is N . Then Sum of digits of N is :

20. If $x \neq 0, y \neq 0, z \neq 0$ and $\begin{vmatrix} 1+2x & 1+x & 1+x \\ 1+2y & 1+3y & 1+y \\ 1+2z & 1+2z & 1+4z \end{vmatrix} = 0$, then the value of $x^{-1} + y^{-1} + z^{-1} + 9$ is

21. If $\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} \geq 0$, where $a, b, c \in \mathbb{R}^+ - \{0\}$, then $\frac{a+b}{c}$ is

22. If $a_1, a_2, a_3, 5, 4, a_6, a_7, a_8, a_9$ are in H.P. and $D = \begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$, then the value of $21D$ is

(Where $[.]$ represents, the greatest integer function)

23. The absolute value of a for which system of equations, $a^3x + (a+1)^3y + (a+2)^3z = 0$,
 $ax + (a+1)y + (a+2)z = 0$, $x + y + z = 0$, has a non-zero solution is:

24. Consider the system of equations

$$x + y + z = 4$$

$$2x + y + 3z = 6$$

$$x + 2y + pz = q$$

Let L denotes the value of p if the system of equations has no solution. and M denotes the value of q if the system of equations has infinite solutions.

Find the unit digit of $(L^2 + M^2)$.

25. If 3^n is a factor of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ {}^nC_1 & {}^{n+3}C_1 & {}^{n+6}C_1 \\ {}^nC_2 & {}^{n+3}C_2 & {}^{n+6}C_2 \end{vmatrix}$ then the maximum value of n is.....

26. Let the matrix A and B be defined as $A = \begin{bmatrix} 3 & 2 \\ 2 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 7 & 3 \end{bmatrix}$. If $\det(2A^9 B^{-1}) = -2$,

then find the number of distinct possible real values of α .

27. Let A_n and B_n be square matrices of order 3, which are defined as

$$A_n = [a_{ij}] \text{ and } B_n = [b_{ij}] \text{ where } a_{ij} = \frac{2i+j}{3^{2n}} \text{ and } b_{ij} = \frac{3i-j}{2^{2n}} \text{ for all } i \text{ and } j, 1 \leq i, j \leq 3.$$

$$\text{If } l = \lim_{n \rightarrow \infty} \text{Tr.} (3A_1 + 3^2A_2 + 3^3A_3 + \dots + 3^nA_n)$$

$$\text{and } m = \lim_{n \rightarrow \infty} \text{Tr.} (2B_1 + 2^2B_2 + 2^3B_3 + \dots + 2^nB_n),$$

then find the value of $(l + m)$.

[Note : Tr. (P) denotes the trace of matrix P.]

Matrix Match Type :

28. Consider a square matrix A of order 2 which has its elements as 0, 1, 2 and 4. Let N denote the number of such matrices.

Column - A

Column - B

- | | |
|---|--------|
| (A) Possible non-negative value of $\det(A)$ is | (P) 2 |
| (B) Sum of values of determinants corresponding to N matrices is | (Q) 4 |
| (C) If absolute value of $(\det(A))$ is least, then possible value of $ \text{adj}(\text{adj}(\text{adj } A)) $ | (R) -2 |
| (D) If $\det(A)$ is algebraically least, then possible value of $\det(4A^{-1})$ is | (S) 0 |
| | (T) 8 |

Subjective Type Questions:

29. If the system of equations $x = cy + bz$, $y = az + cx$ and $z = bx + ay$ has a non-zero solution and at least one of a , b , c is a proper fraction, prove that $a^2 + b^2 + c^2 < 3$ and $abc > -1$.

30. $\Delta = \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$, Show that minimum value of Δ is $27a^2b^2$. Given ab is constant.

31. If t is real and $\lambda = \frac{t^2 - 3t + 4}{t^2 + 3t + 4}$, then find number of solutions of the system of equations $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + \lambda z = -3$ for a particular value of λ .

ANSWER KEY OF CAPS-7

1. (B)	2. (C)	3. (A)	4. (D)	5. (B)
6. (A)	7. (A)	8. (C)	9. (D)	10. (ABC)
11. (A)	12. (D)	13. (A)	14. (B)	15. (D)
16. (A)	17. (9)	18. (4)	19. (2)	20. (3)
21. (2)	22. (50)	23. (1)	24. (6)	25. (3)
26. (1)	27. (21)	28. (A) P, Q, T ; (B) S; (C) P, R ; (D) R]	31. (1)	